



Contents lists available at ScienceDirect

Journal of Experimental Child Psychology

journal homepage: www.elsevier.com/locate/jecp



The development of digital ownership in children

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ARTICLE INFO

Article history:

Received 26 January 2022

Revised 28 June 2022

Available online 5 August 2022

Keywords:

Development

Digital ownership

Digital piracy

Morality

Ownership

Pre-registered

ABSTRACT

The current study examined whether, for children and adults, behaviors that are considered violations of property rights in the case of physical property are likewise viewed as violations in the case of digital property. In this preregistered study ($N = 156$), 5- to 10-year-old children and adults heard a story about a person who downloaded a digital file (e.g., an e-book that she did not own) onto her personal computer (digital) versus a person who put a physical item (e.g., a book that she did not own) into her bag (physical). Participants were asked to evaluate how okay each behavior is. We found that from 5 years of age children evaluated taking a physical object more negatively than downloading a digital file, which was also the pattern observed in adults. Furthermore, by 9 years of age children were equivalent to adults in their evaluation of downloading digital files. The current study has implications for the development of ownership in children.

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Introduction

Digital devices are ubiquitous in modern children's daily life. By 3 years of age, a majority of U.S. children engage with a tablet PC or smartphone, and from 5 to 8 years of age, more than half of children have experience with a computer (Pew Research Center, 2020). With new technology come new moral puzzles that challenge traditional concepts of ownership. Compared with traditional ownership over physical objects, ownership in the digital world presents less clear boundaries between owners (or creators) of a digital file and its permitted users. For example, people commonly download a wide

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range of digital files, including music, movies, e-books, software, comics, photos, and video games—even when they do not own such files and doing so violates existing laws (Bilton, 2012). These digital copyright infringements have made it difficult for content creators and owners to get credit for their hard work. However, despite the importance of digital property rights, little is known regarding the developmental origins of understanding digital property.

Understanding ownership of physical objects emerges early in childhood (see Nancekivell et al., 2019, for a review). For instance, by around 2 years of age, children start to use possessive pronouns (e.g., mine, yours) in peer interactions, demonstrating an early understanding of invisible links between objects and owners (Fasig, 2000; Hay, 2006). From 3 to 5 years of age, children develop inferences concerning ownership such as a first possession heuristic in which the first individual who possessed an object is its owner (Friedman & Neary, 2008; Friedman et al., 2013).

Children's understanding of intellectual property begins to emerge around 5 years of age. For example, by 5 or 6 years, children identify the originator of an idea as its owner and evaluate those who plagiarize more negatively than those who do not, showing an understanding of intellectual property rights violations (Olson & Shaw, 2011; Shaw et al., 2012; Shaw & Olson, 2015). Digital property is similar to intellectual property in that (a) both are intangible and (b) in both cases violations could seem harmless because they do not involve physical loss or harm (e.g., Silver & Shaw, 2018). However, digital downloading also differs in important ways from intellectual property theft. For instance, intellectual property is often a product of its owner's work or effort, whereas digital property does not necessarily reflect its owner's work (e.g., an owner purchases a music file created by someone else). In addition, digital property is multiplicable, and thus it is theoretically possible to have multiple owners in some cases. In contrast, intellectual property is exclusive in nature, assuming one owner (or one group) that can exclusively take credit. Furthermore, in contrast to intellectual property theft, digital ownership violation most typically harms a faceless individual or conglomerate in a digital space, potentially contributing to a failure to recognize the victim (Yu, 2012).

Although studies on intellectual property provided important insights, it is unknown how children think about another intangible property important in our daily life, that is, digital ownership. We are aware of only a few studies that investigated children's understanding of the ownership implications of digital information. By 6 or 7 years of age, children evaluated tracking the location of an object with a mobile GPS device more negatively when a non-owner did the tracking than when an owner did the tracking (Gelman et al., 2018). Furthermore, with increasing age, children evaluated accessing a digital version of another person's private photograph more negatively (Gelman et al., 2021), showing children's growing understanding of digital privacy. Overall, these prior studies suggest that children understand that ownership rights extend to digital information such as tracking the location of others' belongings or accessing a digital copy of others' private photographs. However, these studies did not assess how children evaluate ownership of digital property per se or specifically how behaviors that are considered violations of property rights in the case of physical property compared with comparable behaviors involving digital property.

The current study examined how children and adults evaluate downloading files that one does not own—a behavior that, for physical property, would constitute an ownership violation. This behavior is hypothesized to evoke competing intuitions among adults. On the one hand, a fundamental principle of ownership is that one should not take or use something that is owned by someone else without permission because only the owner has the right to control the item. On the other hand, for digital files (in contrast to physical items), downloading creates a duplicate and thus need not deprive the owner of the owned item. This tension between ownership rights, on the one hand, and the relative lack of harm due to the owner retaining the original, on the other, can be seen in the existence of copyright law (e.g., Digital Millennium Copyright Act) in the United States that prohibits digital copyright infringement as well as rampant online piracy that generally goes unpunished (Johnson, 2020). For example, in a qualitative study (Yu, 2012), although adult participants acknowledged that digital piracy is illegal, they considered it not to be serious or even judged it favorably. This tension provides an opportunity to examine how children conceptualize digital ownership and which of the two competing intuitions is dominant at different ages.

The current study aimed to address two questions. First, does a behavior that constitutes an ownership violation for physical objects also do so for digital property in children? Second, when

do children show an adult-like evaluation in their digital ownership understanding? We focused on the age group of 5- to 10-year-olds given other research showing important changes in children's reasoning about digital ownership during this period (Gelman et al., 2018, 2021) and because previous research has shown that by 5 years of age children extend ownership rights to intangible items such as ideas (Olson & Shaw, 2011). We tested adults as a comparison group in order to examine when children start to show adult-like reasoning. To our knowledge, this is one of the first studies to investigate the developmental trajectory of digital ownership concepts.

The current study presented participants with a series of scenarios in which a hypothetical character either downloaded a digital file from a shared computer onto her personal computer (digital) or transferred a physical object from alongside a shared computer into her bag (physical). Then children were asked to evaluate the degree to which the behavior is acceptable. We specified that the item was on a shared computer (digital) or next to a shared computer (physical) to ensure that participants' judgments focused on ownership of the property rather than ancillary rights (e.g., to enter the room or touch the computer). Our scenarios provided minimal information to avoid cuing children to use certain factors in their judgments. This design permitted children to apply default interpretations to what constitutes acceptable behavior toward physical (versus digital) property in contexts that are otherwise identical, somewhat ambiguous, and commonly encountered. Thus, we did not specify the owner of the property and did not specify whether the item had been left intentionally or accidentally.

Based on the existing literature on ownership of physical objects (Kanngiesser & Hood, 2014; Nancekivell & Friedman, 2017; Rossano et al., 2011), we expected that children would evaluate the violation of physical ownership negatively, although this tendency may become stronger with age. More specifically, given that the vignettes involved an ambiguous setting in which the owner was not specified and it was unclear whether the item had been left intentionally or accidentally, we expected that older children might be more adept at inferring ownership from the ambiguous context and thus evaluate the violation of physical ownership more negatively compared with younger children.

The critical question was how children would evaluate scenarios in the digital ownership context. One hypothesis was that children would evaluate downloading a digital file negatively due to the primacy of ownership—specifically, the principle that owners have privileged control over items that non-owners do not share. If obtained, these results would suggest that children apply the ownership rule that is typically applied to physical objects to digital cases as well.

An alternative hypothesis was that children would evaluate downloading a digital file positively or neutrally at worst. This possibility is plausible because digital files are duplicable and thus perceived harm should be less severe in digital cases. Findings consistent with this hypothesis would suggest that children's sense of ownership over physical objects is not automatically applied to digital property, implying that children consider digital files as an exception to ownership principles.

Adults were also included as a comparison group. Although prior qualitative research suggests that adults judge that downloading a digital file would be more acceptable than taking a physical object (Yu, 2012), we are unaware of any study directly comparing these two cases while controlling for other factors. We were interested to learn how children's responses would compare with those of adults as well as whether/how children's responses would change with age. As noted above, we anticipated that children's judgments would likely be more negative with age, reflecting an increasing sensitivity to ownership rights in this more ambiguous and challenging context (i.e., a shared space where the owner is not present). However, it was not clear whether this anticipated age change would be comparable for digital and physical items.

Method

Participants

Our final sample included 96 5- to 10-year-old U.S. children ($M = 95.83$ months, range = 60–129; 15 5-year-olds, 16 6-year-olds, 17 7-year-olds, 16 8-year-olds, 16 9-year-olds, and 16 10-year-olds; 51 female and 45 male) and 60 adults ($M = 37.7$ years, range = 24–66; 23 female and 37 male). Adult

participants in the United States were tested through Amazon Mechanical Turk (MTurk) before the pandemic in October 2019.

Prior to the COVID-19 pandemic, children were tested at a museum in the United States ($n = 37$) from February to early March 2020. During the pandemic, children were tested via an online Zoom meeting platform ($n = 59$) from October 2020 to February 2021. The experimental stimuli and procedure were nearly identical between the two testing platforms (see “Experimental design and procedure” section for differences). Furthermore, statistical analysis confirmed that the testing platforms did not bias our results significantly ($p > .34$).

An additional 21 children were excluded because they failed to use the Likert scale properly during practice trials ($n = 7$) or because they answered at least one of the comprehension questions about digital and physical formats incorrectly ($n = 14$).

Our child sample was reported by parents to be as follows: White (61%), multiracial (10%), Black (5%), Hispanic (5%), and Asian American (1%), with 17% indicating “other” or not reporting. Our adult sample self-reported as follows: White (83%), Black (7%), Asian American (3%), multiracial (3%), Hispanic (2%), and Native Hawaiian (2%). We preregistered our hypotheses and analyses before data collection in both the adult study (<https://aspredicted.org/7w34w.pdf>) and the child study (<https://aspredicted.org/www57e.pdf>).

Experimental design and procedure

After parents provided consent, children were brought to a table with the study apparatus (when they participated at a museum) or researchers scheduled an online testing session in which children participated with an electronic device (when they participated online).

During a *practice phase*, children were asked to indicate how okay or not okay it is to give a desirable gift (e.g., Lego) or an undesirable gift (e.g., broken cup) to another child. Children were shown a visual representation of a 5-point Likert scale (e.g., two thumbs down for totally not okay) to indicate their judgments (from -2 being totally not okay to $+2$ being totally okay). This visual aid was used throughout the test phase. All children included in the final sample were able to use the Likert scale properly (e.g., reported that it is not okay to give an undesirable gift).

During the subsequent *introduction phase*, children were introduced to both digital and physical items. Specifically, in an introduction to a digital item, the experimenter demonstrated how downloading works by showing that a music file from the first computer can be downloaded to the second computer. Similarly, in an introduction to a physical item, the experimenter showed a music CD and demonstrated putting it in a bag by moving it from a box to the bag. (When tested online, this phase was replaced by showing a video of this introduction phase.)

During the following *test phase*, children heard eight scenarios in total: 2 formats (digital or physical) $\times$ 4 item types (book, photo of nature, drawing, or game). In one block, the four items were presented in a digital format. In the other block, children heard the same four items in a physical format. (The order of the blocks was counterbalanced.) For example, in one of the scenarios in a block with digital items (Fig. 1A), children heard the following: “[X] finds that a book was saved on a shared computer in the room. [X] is the only person in the room. No one will know if [X] downloads the book on her computer. Is it okay or not okay for [X] to download the book on her computer? How okay or not okay is it?” In the physical version, the story was identical except that the item was shown as a physical object (e.g., “[X] finds that a book was left next to a shared computer in the room. [X] is the only person in the room. No one will know if [X] puts the book in her bag. Is it okay or not okay for [X] to put the book in her bag? How okay or not okay is it?”; see Fig. 1B). Thus, children were asked how okay or not okay it is to download the item (digital) or to put it in the protagonist’s bag (physical). At the end of each scenario, children were asked to explain their answer (“Why do you think it is okay or not okay?”).

After the test phase, to check children’s understanding of the different formats, the study concluded with two comprehension questions each for a digital item and a physical item, respectively (thereby resulting in four questions in total), assessing (a) whether children remembered where the item was at first (e.g., “Where was the book at first? Was it on the table or was it in her bag at first?”) and (b) whether they understood that a physical item cannot exist in two places at the same time, whereas

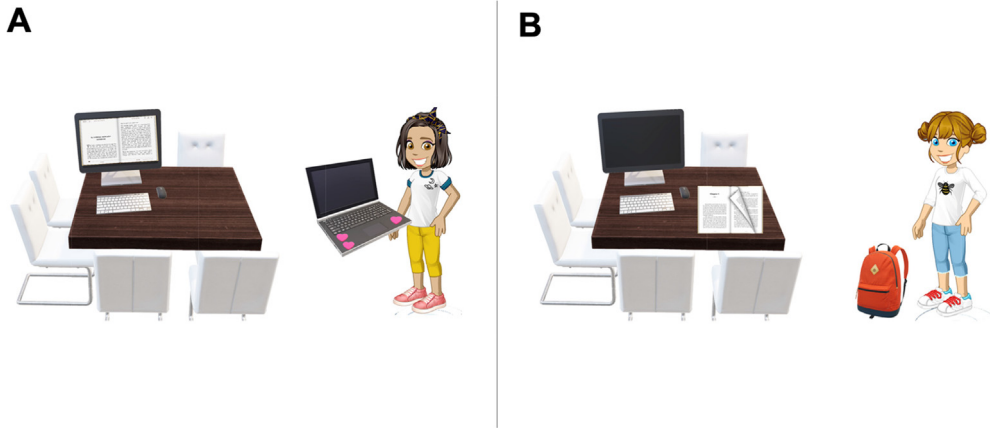


Fig. 1. Examples of stimuli shown to children in a version with a digital item (A) and a physical item (B).

a digital item can exist on both computers. (For example, to check comprehension of digital formats, the experimenter asked, “If [X] downloads the book on her computer, would it be only on her computer, only on the first computer, or on both computers?”) All children retained for the final sample were correct in answering these questions.¹ We counterbalanced the order of blocks (digital or physical), item types within each block (book, photo of nature, drawing, or game), and protagonist characters.

Adults read the same scenarios with a few minor differences. First, adults did not receive the practice and introduction phases and did not receive the comprehension questions. Second, adults read stories via Qualtrics without visual representations. Third, adults were presented with an additional two items—music and movie—which were excluded from the child study because pilot testing revealed that children were not familiar with a cassette tape or a videotape (the physical instantiations of these items). Lastly, for adults, the same item did not appear twice (e.g., a book appeared only once as either a digital or physical item). Therefore, adults read six scenarios in total (music, movie, book, photo of nature, drawing, and game) in orders randomized by Qualtrics, with three items presented in a digital format and the other three items presented in a physical format. In addition, the types of items shown in a digital versus physical format were counterbalanced.

Data coding and analyses

Responses from adults were recorded by Qualtrics. Responses from children were recorded by the experimenter and were later entered into a spreadsheet. Answers to open-ended questions from the 84 child participants who consented to audio recording were later transcribed and independently coded by two coders. Answers to open-ended questions from all 60 adult participants were also coded by two independent coders. Average Cohen’s kappas were 0.79 in the child study and 0.84 in the adult study. We predetermined the categories for coding explanations based on pilot testing with adults (see Table 1).

All statistical analyses were conducted with R statistical software (R Version 4.1.1; [R Core Team, 2021](#)). We analyzed ratings with a linear mixed model (LMM) using the R package “lme4” ([Bates et al., 2015](#)) and analyzed justifications with a generalized linear mixed model (GLMM) using the R package “glmmTMB” ([Brooks et al., 2017](#)). Our full LMMs on ratings included format type, item type, children’s age in months (as a continuous variable in the child study), and all interactions among the variables as fixed effects and participant identity (ID) as a random effect. Our full GLMMs on justifica-

¹ The pattern of results was identical even when we included responses from the 14 children who failed the comprehension check questions.

Table 1
Coding categories for open-ended questions.

Category	Criterion	Examples
Ownership	Mentioning the ownership of objects (or lack of ownership) to explain why it would be okay or not okay.	"That was someone else's property." "She's taking away from the owner."
Legality	Mentioning whether it is legal or illegal to explain why it would be okay or not okay.	"It's stealing." "It's illegal."
Public nature	Mentioning public sharing or public access of the property (or the private nature of the item) to explain why it would be okay or not okay.	"The game is for every-one." "It's in a public space."
Duplicability	Mentioning duplicability or originality to explain why it would be okay or not okay.	"It will still be there even if she downloads it." "They are taking the only copy of a thing and depriving others of using it."
Harmlessness	Mentioning the harmlessness or harmfulness of the action to explain why it would be okay or not okay.	"It wouldn't impact anyone." "Sammy isn't hurting anyone by doing this."
Morality	Mentioning whether it is right or wrong based on a general moral standard to explain why it would be okay or not okay.	"I don't think that is wrong." "It might be a little immoral."
Item value	Mentioning either the personal or monetary value of the item (either how valuable it is or how valueless it is) to explain why it would be okay or not okay.	"The photo doesn't seem to be personal." "It costs money."
Possibility of being caught	Mentioning the possibility of being caught (or the lack of possibility of being caught) to explain why it would be okay or not okay.	"No one will know." "Someone will find out who took it."
Conditional response	Mentioning a decision criterion, while putting off his or her decision about why it would be okay or not okay.	"It depends on whether it's a copyrighted song or not." "If it is open source, it's okay. If it's not, it's IP [intellectual property] theft."

Note. Example sentences in this table were extracted from pilot testing with adults.

tions included format type, coding category, and the interaction between the variables as fixed effects and participant ID as a random effect.

We first compared the full model with a null model that included only participant ID as a random intercept. If the full model provided a significantly better fit to the data than the null model, we then conducted hypothesis-driven tests to examine the role of individual predictors by sequentially dropping them from the full model and assessed changes in model fit using likelihood ratio tests (LRTs). We also conducted one-sample *t* tests (two-tailed) to examine whether ratings differed significantly from a neutral valence of 0.

For the sake of brevity, we do not report the statistics from comparisons between full and null models. All our full models provided a significantly better fit to the data than the null models (*ps* <.001).

All data and protocols are available through the Open Science Framework (<https://osf.io/3g2j4/>).

Results

Adults

We assessed whether adults' ratings were influenced by format type (digital or physical) and item type (book, photo, drawing, game, music, or movie). We found a significant main effect of format type, LRT, $\chi^2(1) = 194.2$, $b = -1.46$, $SE = 0.09$, $p < .001$, whereas there were no main effects or interaction effects involving item types (LRT, all *ps* >.07). That is, across different items, adults rated taking a physical item ($M = -1.30$, $SD = 0.88$) more negatively than downloading a digital item ($M = 0.16$, $SD = 1.33$).

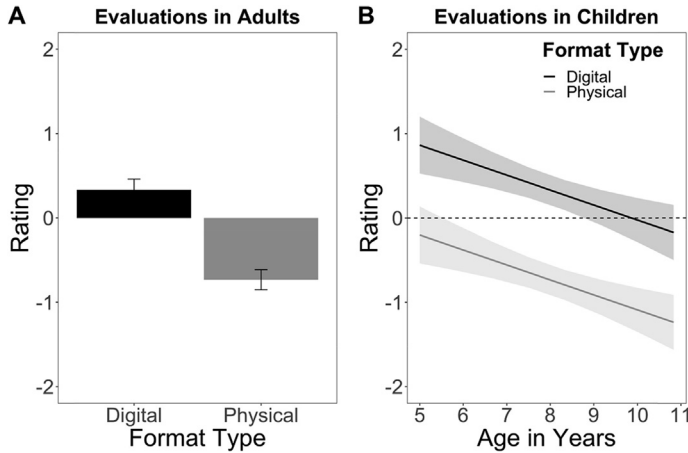


Fig. 2. (A) Adults' ratings of downloading digital items versus taking physical items. (B) Estimated ratings of downloading digital items versus taking physical items in children. Error bars represent 95% confidence intervals.

Further one-sample t tests revealed that their evaluation of taking a physical item differed significantly from the neutral valence of 0, $t(179) = -19.74$, $p < .001$, whereas that of downloading a digital item did not differ significantly, $t(179) = 1.62$, $p > .10$ (Fig. 2A). The results indicate that adults evaluated taking a physical item negatively, whereas they viewed downloading a digital item in a neutral manner.

To assess how adults' justifications differed by format type and coding category, we ran a full GLMM on coding with main and interaction effects between format type and category, which revealed a significant interaction between format type and category, LRT, $\chi^2(8) = 112.24$, $p < .001$ (Fig. 3A). To unpack the interaction, we ran separate analyses on each category. We found that adults were more likely to refer to ownership and legality in scenarios involving physical items (versus digital) items, LRT, $\chi^2(1) = 46.75$, $b = 2.03$, $SE = 0.34$, $p < .001$ and $\chi^2(1) = 7.70$, $b = 0.92$, $SE = 0.34$, $p < .01$, respectively. In contrast, they were more likely to refer to public nature, duplicability, harmlessness,² and conditional response in scenarios involving digital (versus physical) items, LRT, $\chi^2(1) = 38.94$, $b = -3.33$, $SE = 0.85$, $p < .001$ for public nature, $\chi^2(1) = 17.39$, $b = -3.03$, $SE = 0.93$, $p < .001$ for duplicability, and $\chi^2(1) = 7.46$, $b = -2.12$, $SE = 0.90$, $p < .01$ for conditional response. Other categories were not significantly different between format types ($ps > .10$).

Children

Next, we assessed whether children's ratings were influenced by format type (digital or physical), item type (book, photo, drawing, or game), and age. We found a significant main effect of format type, LRT, $\chi^2(1) = 214.89$, $b = -1.06$, $SE = 0.07$, $p < .001$, and a significant main effect of age, LRT, $\chi^2(1) = 12.20$, $b = -0.01$, $SE = 0.004$, $p < .001$. There were no significant main effects or interaction effects involving item type (LRT, all $ps > .06$). That is, like adults, children rated taking a physical item ($M = -0.73$, $SD = 1.19$) more negatively than downloading a digital item ($M = 0.33$, $SD = 1.26$) (Fig. 2B). Furthermore, with increasing age, they were more likely to judge both behaviors negatively ($b = -0.01$, $SE = 0.004$, $p < .001$). Adding the testing platforms (museum or online) or the block order (digital items first or physical items first) did not provide a better fit compared with our model, LRT, $\chi^2(5) = 5.65$, $p > .34$ and $\chi^2(5) = 6.48$, $p > .26$, respectively.

To assess the age at which children's evaluations of taking physical items became significantly negative, we computed the age point when confidence intervals no longer included a neutral valence of 0.

² We were unable to provide statistics for harmlessness because the model did not converge, potentially because there were no instances in which participants referred to harmlessness in physical items (see Fig. 3A).

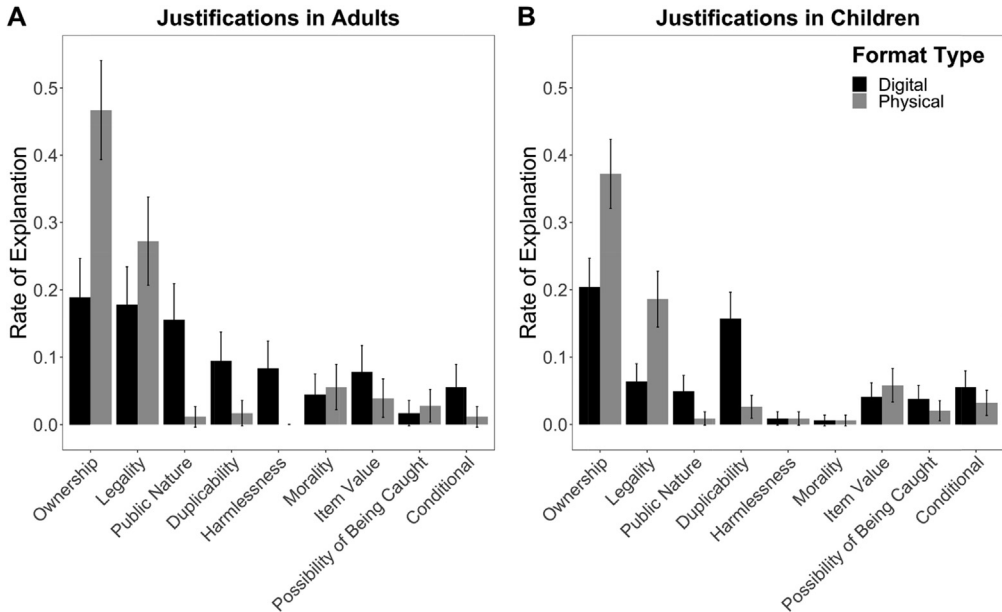


Fig. 3. Adults' justifications (A) and children's justifications (B) of their ratings. Error bars represent 95% confidence intervals.

The results were that from 67 months of age (5.6 years), children started to evaluate taking physical items negatively. Similarly, we computed the age point when confidence intervals of digital items started to change from positive to neutral levels. The findings were that, from 106 months (8.8 years), their confidence intervals started to overlap with 0, suggesting that around 9 years of age children (like adults) started to evaluate downloading digital items in a neutral manner.

Next, we ran a full GLMM on coding with main and interaction effects between format type and category. Further analysis revealed a significant interaction between format type and category, LRT, $\chi^2(8) = 105.01$, $p < .001$ (Fig. 3B). To unpack the interaction, we ran separate analyses on each category. We found that children, like adults, were more likely to refer to ownership and legality in scenarios involving physical (versus digital) items, LRT, $\chi^2(1) = 35.12$, $b = 1.25$, $SE = 0.22$, $p < .001$ and $\chi^2(1) = 42.64$, $b = 2.28$, $SE = 0.41$, $p < .001$, respectively. In contrast, they were more likely to refer to public nature, duplicability, and the possibility of being caught in scenarios involving digital (versus physical) items, LRT, $\chi^2(1) = 12.18$, $b = -1.95$, $SE = 0.66$, $p < .001$, $\chi^2(1) = 55.66$, $b = -2.94$, $SE = 0.50$, $p < .001$, and $\chi^2(1) = 5.79$, $b = -2.13$, $SE = 1.03$, $p < .05$, respectively. There were no significant differences between format types in other categories ($ps > .11$).

Discussion

The current study is one of the first studies to directly investigate children's understanding of digital ownership. Specifically, we asked both 5- to 10-year-old children and adults to judge taking a physical object versus downloading the digital equivalent. This study permits insights into whether/how young children apply ownership principles to digital items, which for adults raise competing intuitions. Moreover, by comparing digital and physical items, we provide a tightly controlled design, such that downloading of digital items can be compared with taking physical items when context, item content, characters, and information regarding ownership are kept constant.

Our results suggest that children as young as 5 years already exhibit sensitivity to different formats (digital versus physical). Like adults, children across ages viewed downloading digital items as more acceptable than taking physical items. Moreover, around 9 years of age, children's evaluations of

downloading digital items dropped from positive to neutral, consistent with the interpretation that children reached adult-like evaluations of digital ownership around this age. In addition, we explored the possibility that the neutral ratings in digital scenarios are due to a potential bimodal distribution (e.g., half the participants giving negative ratings and half giving positive ratings). However, visualization of rating frequencies revealed that neither adults nor children showed a bimodal distribution in ratings of digital scenarios (see Section 1 of the online [supplementary material](#) for details).

Another notable finding is that children started to judge taking a physical item negatively from 5.5 years of age, not from the youngest possible age we tested. This result contrasts with earlier findings that even 2- and 3-year-olds understand ownership of physical objects and protest when ownership norms are violated (Kanngiesser & Hood, 2014; Rossano et al., 2011; see Nancekivell et al., 2019, for a review). However, unlike previous research, the current scenarios provided minimal cues regarding ownership (i.e., the owner was never named or even present, and the context was a shared room) while keeping these cues constant between digital and physical scenarios.³ In addition, it was unclear whether an item was left intentionally or accidentally, which could be a critical factor in determining how likely it was that the owner would come back to retrieve the item and whether it is okay to take the item. Hence, we suspect that the developmental changes in the current study reflect children's growing skills at inferring ownership-related cues that are not explicitly provided but must be inferred.

Children's justifications for their judgments further provide important insight into their reasoning. We found that children more often referred to ownership and legality in physical scenarios, whereas they referred to duplicability and public nature almost exclusively in digital scenarios. Adults also showed these patterns, suggesting that children (who presumably have less digital experience than adults) already exhibit reasoning similar to that of adults. Furthermore, our finding that both children and adults were more likely to refer to duplicability and public nature in digital (versus physical) items hints that the low level of perceived harm might have influenced their judgments, resulting in more positive evaluations in digital scenarios.

The current study provides new evidence regarding children's understanding of intangible property. As noted in the Introduction, by 5 or 6 years of age, children show an understanding of intellectual property rights violations (Olson & Shaw, 2011; Shaw et al., 2012; Shaw & Olson, 2015). It is perhaps not surprising that our findings suggest that digital ownership understanding appears to reach adult-like reasoning later in development than intellectual ownership. This may be because unlike intellectual property, digital properties are multiplicable and it is relatively difficult to recognize or identify the victim in a digital space. It is, however, important to note that the current study measured digital ownership differently from previous research on intellectual ownership. For example, prior studies (Olson & Shaw, 2011; Shaw & Olson, 2015) contrasted a scenario in which intellectual property rights were violated with a scenario in which such rights were not violated, and children were asked to judge both scenarios. In contrast, the current study compared two similar scenarios, varying whether the key item was digital or physical. This comparison was made specifically to examine whether behaviors considered violations of property rights in the case of physical property are likewise viewed as violations in the case of digital property. Therefore, the direct comparison of the current findings with results from intellectual property violations should be interpreted with caution.

In addition, the current work extends prior work on digital ownership. For example, a recent study (Gelman et al., 2021) showed that, with age, children from 5 to 10 years judged accessing another person's private digital photo (e.g., a messy room) more negatively than accessing a public digital photo (e.g., a rainbow). That work demonstrated children's understanding of one of the violations in digital ownership—accessing others' private digital property. However, it was unclear how children would judge another common violation—downloading others' digital property. Unlike previous work, we compared children's judgments of downloading digital files with taking physical items. This comparison allowed us to examine whether children apply the ownership rule that is typically applied to physical objects to digital cases as well. Overall, both the prior work and the current work suggest that

³ We ran a supplementary study with adults to address a possibility that participants inferred ownership in physical cases but not in digital cases. In this supplementary study, we specified an owner of the target item in both physical and digital scenarios. Even though the owner was specified in both scenarios, adult participants still evaluated taking a physical item more negatively than downloading a digital item. Refer to Section 2 of the online [supplementary material](#) for more details.

children's sense of digital ownership becomes stronger from 5 to 10 years of age given the findings that children judge accessing a digital photo (Gelman et al., 2021) or downloading a digital file (current study) more negatively with increasing age. The current study contributes to the literature on the development of digital ownership by examining children's understanding of downloading digital files.

Our findings do not necessarily imply a principled distinction between digital and physical property. Instead, digital property and physical property differ in their formats and affordances. If these differences were eliminated, the difference in evaluations may likewise disappear (e.g., if downloading the file were to remove a digital file from the owner's computer, if a thief were to make and leave a duplicate when stealing an object from its owner; but see Hood & Bloom, 2008). Although the current research differentiated digital property from physical property based on co-occurring attributes prevalent in (current) real life, future research could tightly control these aspects between digital and physical property. Moreover, our findings do not necessarily suggest that the development of digital ownership is completed by 9 or 10 years of age. For instance, it is possible that understanding of digital ownership is susceptible to changes during adolescence in which the digital world and video-gaming become a major part of social life (Pew Research Center, 2015).

We should address potential limitations in the study. First, one could argue that digital downloading is less harmful than physical taking and that this is why we obtained differences between digital and physical items on this task. However, we believe that loss from digital downloading is never trivial. For example, global piracy of movies and TV shows costs the U.S. economy at least \$29.2 billion annually in lost revenue and at least 230,000 job losses (McCarthy, 2019). Indeed, we think that the *subjective* perception that digital downloading is harmless may lie at the heart of our findings. Participants' sense that the digital scenario is not a serious transgression may explain the prevalence of digital piracy outside the lab context.

Second, it is possible that children in our study did not understand the nature of digital items or how downloading works. However, we believe that this is unlikely for three reasons. First, we had an introduction phase during which children were shown how downloading worked between two computers. Second, all the children included in the final sample passed the comprehension questions asking whether they understood the nature of each format. Third, children's explanations for their judgments were very similar to those of adults, demonstrating that children did not simply provide similar ratings as adults but also engaged in similar lines of reasoning.

The current study provides the experimental evidence on children's understanding of digital ownership, and as such it opens up numerous interesting new research directions. Future research could investigate whether knowing that the owner of a digital file is an individual or a company sways children's perceptions of digital ownership. Furthermore, the current study examined one specific behavior in digital ownership, that is, downloading files. However, future research could test other behaviors that would constitute digital ownership violations such as distributing another person's digital file for making money or using it without giving credit to its creator. Lastly, it would be important to study whether emphasis on potential harm from downloading files could change children's evaluations. Together, these future studies would contribute to understanding how children conceptualize digital ownership and how to improve awareness of digital piracy in the real world.

Acknowledgments

This research was funded by a Rackham Research Grant awarded to Y-e.L. We thank Tania Dhaliwal, Allison Sparpana, Claudia Rosas, and Isabelle Kildee for their help with data collection and thank Alaa Al-Kahalalah, Alyssa Knecht, Isabelle Kildee, Kat Black, and Valerie Umscheid for their help with coding data.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jecp.2022.105519>.

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